Ziyi Feng

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Haidian District, Beijing, China

EDUCATION

University of Science and Technology Beijing (Project 211) Beijing, China
B.S. in Information Security (Elite Program) Sep. 2023 – Present

- **Cumulative GPA:** 3.77 / 4.0 (88.9/100) **Ranking:** 5 / 73 (6.85%)
- **Key Courses:** Computer Network Experiments (100), Principles of Compilers (99), Mathematics Fundamental for Information Security (98), Modern Cryptography (97), Introduction to Information Security (97), Physical Education I (95), Principle of Computer Organization (93), Data and Algorithm (90).

RESEARCH EXPERIENCE

DFA Mechanism in Neural Networks applied to Transformer Architectures Sep. 2025 – Present
Visiting Student, WICA Lab, Westlake University **Supervisor:** Prof. Haibin Ling (IEEE Fellow)

- Modified the MLP module in Vision Transformer (ViT) architectures by replacing standard Backpropagation (BP) with Direct Feedback Alignment (DFA).
- Improved model training speed and reduced memory footprint while maintaining accuracy for image classification tasks.
- **Publication:** Paper submitted to NeurIPS 2026.

Research on LLM-as-a-Judge in Large Context Language Models Nov. 2025 – Present
Individual Project, USTB **Supervisor:** Dr. Bowen Xing

- Introduced functional roles (Scorer, Referee, Auditor, Coach) to evaluate system defects in long-context reasoning, consistency, and explainability.
- Proposed a unified analysis framework for long-text evaluation focusing on context modeling and decision consistency.
- **Publication:** Review paper submitted to TPAMI 2026.

Medical Large Language Model Data Alignment & Fine-tuning Mar. 2025 – Jun. 2025
Core Member, USTB **Supervisor:** Dr. Bowen Xing

- Used LLM APIs to clean and process unstructured patient data, utilizing NLP to extract medical indicators.
- Standardized medical values into fixed formats, improving data readability and efficiency for medical information management.

COMPETITION EXPERIENCE

Mathematical Contest in Modeling (MCM/ICM) Jan. 2025 – Feb. 2025
Honorable Mention (International)

- Developed a K-means++ clustering model to classify global cybercrime risk levels based on standardized crime rates and internet penetration.
- Built a Generalized Linear Model (GLM) using Poisson distribution to analyze socioeconomic factors affecting crime rates.

National College Students' Career Planning Competition Dec. 2025 – Mar. 2026
Gold Award (Beijing Municipal Level)

- Awarded the Beijing Municipal Gold Award in the Growth Track (Higher Education Group) with a top 1% performance.

China Undergraduate Mathematical Contest in Modeling (CUMCM) Aug. 2025 – Sep. 2025
Second Prize (Beijing Division)

- Implemented a Particle Swarm Optimization (PSO) framework to optimize UAV deployment strategies for smoke screen interference.
- Integrated Simulated Annealing (SA) to improve global convergence and stability, ensuring optimal concealment duration.

National Undergraduate Statistical Modeling Contest Mar. 2025 – May. 2025
Third Prize (Beijing Division)

- Designed a Gaussian Process Regression (GPR) model with mixed kernels to analyze flight trajectories, achieving an 89.3% coverage rate for 95% confidence intervals.
- Built a Deep Feedforward Network (DNN) with a dual-hidden layer architecture (128-256) for adaptive prediction in dynamic environments.

The 18th Software System Security Contest Nov. 2024 – Feb. 2025
Third Prize (North China Division)

- Proposed an improved TrAdaBoost algorithm for industrial control malicious traffic detection, addressing data imbalance issues through sample weight adjustment.

HONORS & AWARDS

- Taishan Steel Scholarship (Specialized Scholarship) 2024
- Taihu Talent Dingfei Technology Scholarship (Specialized Scholarship) 2025
- University Freshman Scholarship (Second Prize) 2024
- USTB "Outstanding Merit Student" 2024
- USTB "Outstanding Merit Student" 2025
- USTB Advanced Individual in Social Practice 2024
- **Has accumulated over 10 awards at national, provincial and municipal levels.**

SKILLS

- **Programming & Tools:** Python, C++, LaTeX, Verilog
- **Languages:** English (CET-6), Chinese (Native)